

```
Initial data shape: (5000, 10)
```

	Temperature	Humidity	PM2.5	PM10	NO2	SO2	CO	\
0	29.8	59.1	5.2	17.9	18.9	9.2	1.72	
1	28.3	75.6	2.3	12.2	30.8	9.7	1.64	
2	23.1	74.7	26.7	33.8	24.4	12.6	1.63	
3	27.1	39.1	6.1	6.3	13.5	5.3	1.15	
4	26.5	70.7	6.9	16.0	21.9	5.6	1.01	

	Proximity_to_Industrial_Areas	Population_Density	Air Quality
0	6.3	319	Moderate
1	6.0	611	Moderate
2	5.2	619	Moderate
3	11.1	551	Good
4	12.7	303	Good

```
Missing values per column:
Temperature          0
Humidity             0
PM2.5               0
PM10                 0
NO2                  0
SO2                  0
CO                   0
Proximity_to_Industrial_Areas  0
Population_Density    0
Air Quality          0
dtype: int64
```

```
Shape after removing outliers: (4741, 10)

Dataset columns:
Index(['Temperature', 'Humidity', 'PM2.5', 'PM10', 'NO2', 'SO2', 'CO',
      'Proximity_to_Industrial_Areas', 'Population_Density', 'Air Quality'],
      dtype='object')
```

```
Dataset info:
<class 'pandas.core.frame.DataFrame'>
Index: 4741 entries, 0 to 4999
Data columns (total 10 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Temperature                             4741 non-null   float64
1   Humidity                               4741 non-null   float64
2   PM2.5                                  4741 non-null   float64
3   PM10                                   4741 non-null   float64
4   NO2                                    4741 non-null   float64
5   SO2                                    4741 non-null   float64
6   CO                                     4741 non-null   float64
7   Proximity_to_Industrial_Areas          4741 non-null   float64
8   Population_Density                     4741 non-null   int64
9   Air Quality                            4741 non-null   object
dtypes: float64(8), int64(1), object(1)
memory usage: 407.4+ KB
```



